MIL-L-12997D

18 September 1972
SUPERSEDING
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Amendment 2
20 June 1966

# MILITARY SPECIFICATION

## LINK, CARTRIDGE, METALLIC BELT, 20MM, M12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers one type of metallic belt link (welded or non-welded) intended to hold 20mm ammunition in the form of flexible belts for use in the M39 type 20mm automatic gun.

## 2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | | |
|---|---|---|
| MIL-P-116 | - | Preservation, Methods of. |
| MIL-W-6858 | - | Welding, Resistance: Aluminum, Magnesium, Non-Hardening Steels or Alloys, and Titanium Alloys Spot and Seam. |
| MIL-W-13855 | - | Weapons: Small Arms and Aircraft Armament Subsystems, General Specification for. |
| MIL-W-45223 | - | Welding, Spot, Hardenable Steels. |
| MIL-I-45607 | - | Inspection Equipment, Acquisition, Maintenance and Disposition of. |

STANDARDS

Military

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-109 | - | Quality Assurance Terms and Definitions. |
| MIL-STD-171 | - | Finishing of Metal and Wood Surfaces. |

FSC 1305

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DRAWINGS

U.S. Army Weapons Command

| | | |
|---|---|---|
| D7147028 | - | Link, Cartridge, Metallic Belt, 20MM, M12 |
| D7548085 | - | Link, Cartridge, Metallic Belt, 20MM, M12 (Alternative) |

(Product drawings referenced in this specification form a part of D7147028 or D7548085.)

PUBLICATIONS

U.S. Army Weapons Command

| | | |
|---|---|---|
| P7147028 | - | Packaging Data Sheet for Link, Metallic Belt, 20MM, M12. |
| IEL7147028 | - | Index of Inspection List. |
| IEL7548085 | - | Index of Inspection List. |

(Acceptance Inspection Drawings referenced in this specification form a part of IEL7147028 or IEL7548085.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Material

ASTM E18 - Test for Rockwell Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

* 3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

* 3.2 Material, construction and design. Links shall conform to material, construction and design requirements specified herein, on Drawing D7147028 or D7548085 as applicable, and drawings applicable thereto, and be in accordance with the applicable material, construction and design provisions of MIL-W-13855.



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3.2.1 Heat treatment. Links shall be heat treated to a hardness within the hardness range specified on the applicable drawings. Heat treatment shall be conducted in such a manner that the formation of decarburization is not in excess of the limits specified on the applicable drawings. Testing shall be as specified in 4.5.3.3 and 4.5.3.4.

* 3.2.2 Pull. Links shall be capable of withstanding 600 + 25 pounds tensile load without fracture. When tensile load is applied, the distance X-X as shown on D7147028 or D7548085 shall not exceed 1.945 at load and the bridge shall be tight on the body after test. Testing shall be as specified in 4.5.3.5 or 4.5.3.5.1, as applicable.

* 3.2.3 Twist. Links assembled into belts of 25 links with Government standard 20mm dummy cartridges shall be capable of withstanding a 180° twist with no evidence of partial or complete belt separation. Testing shall be as specified in 4.5.3.6.

* 3.2.4 Cartridge retention. The force required to strip the Government standard 20mm dummy cartridges from links shall not be less than 125 pounds nor more than 300 pounds applied parallel to the longitudinal centerline of the cartridge. Testing shall be as specified in 4.5.3.7.

* 3.2.5 Free hinging. Links assembled into belts of 25 links with Government standard 20mm dummy cartridges shall hinge and fold over without binding, kinking or any interference between the projecting surfaces of the links. Testing shall be as specified in 4.5.3.8.

* 3.2.6 Functioning. Links shall be capable of being assembled with Government standard 20mm cartridges into belts and shall be operated in the normal firing of an M39 type automatic gun without malfunction, cracking, breaking, or separating. Testing shall be as specified in 4.6.6.

* 3.2.7 Welding. Welding of the stress bridge shall be in accordance with MIL-W-45223 or Class B of MIL-W-6858. Testing shall be as specified in 4.5.3.9.

* 3.3 Protective finish. The links shall be finished with finish number 5.3.1.2 or 5.3.2.2 of MIL-STD-171.

3.4 Marking. The links shall be marked in accordance with the marking requirements of MIL-W-13855.

* 3.5 Workmanship. Workmanship shall be in accordance with the workmanship requirements of MIL-W-13855. Cracks, tears, burs and deformations shall not exceed the limits defined in 6.3.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

* 4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

* 4.4 First article inspection. The first article shall be selected from the first inspection lot and submitted for inspection in accordance with contract requirements. The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, the function test specified in 4.6.6 and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

* 4.5.1 Inspection lot. Unless otherwise specified, the formation, size and presentation of inspection lots shall be packaged and applicable samples shall be withdrawn from the packaged links in accordance with MIL-STD-105. Inspection lots shall not exceed 500,000 links per lot. Lots shall be from the same heat of steel and within the limitations of MIL-W-13855.

* 4.5.1.1 Heat treat lot. The heat treat lot shall be limited to one heat steel and shall consist of links heat treated in a batch-type furnace at one charge or in a continuous-type furnace during an uninterrupted period extending over not more than one eight-hour shift.



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4.5.2 Examination. Examination of links and packaging shall be performed as specified herein and in accordance with the applicable examination provisions of MIL-W-13855.

* 4.5.2.1 Classification of defects. Classification of defects and acceptable quality levels (AQL's) for examination in accordance with MIL-STD-105 shall be as specified herein and shall include the following provisions:

a. Unless otherwise specified, AQL's apply to each listed characteristic.
b. Where a Government drawing is listed as the method of inspection, the contractor shall use inspection equipment fabricated in accordance with the applicable drawings. In the event the contractor desires to use alternate equipment which equals or exceeds the accuracy requirements of the specified drawings, it shall be submitted to the Government representative for approval.
c. Where "visual" is specified as the method of inspection of protective coatings, the coating must be complete, uniform in color, free from pits, corrosion, scratches, and worn or bare spots.
d. Where "visual - manual" is specified as the method of inspection of functioning assembly requirements, the assembly shall be visually inspected for completeness and manually operated for functioning requirements specified on the applicable drawing.
e. Inspection for Workmanship shall include examination for missing manufacturing operations and the presence of nicks, burs, cracks, scratches, tool marks, deformations or other signs of poor workmanship.

* 4.5.2.1.1 Link Cartridge Metallic Belt, 20MM, M12 (Drawing D7147028 and D7548085).

| Categories | Characteristics | AQL | Inspection Method |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | | | |
| 101 | Cracks, tears, and deformations (see 6.3). | 0.065 | Visual |
| 102 | Missing or improper weld buttons (for welded link only) | 0.065 | Visual - Manual |
| 103 | Width; end, stress arm (w/o ring) | 0.25 | A8415842, A8415843 |

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| Categories | Characteristics | AQL | Inspection Method |
|---|---|---|---|
| 104 | Max. clearance between bridge or body arms | 0.25 | A7190493 |
| 105 | Min. distance from bridge arms or body arms to vertical center line of link (w/steel plug in link) | 0.25 | B7190405 |
| 106 | Min. distance from tip of flared end of body stress arm (w/ring) to longitudinal center line of link (w/steel plug in link) | 0.25 | B7190405 |
| 107 | Grip on cartridge (min. opening) | 0.65 | C7190032 |
| 108 | Height of bridge tip from bridge base | 0.65 | C7190492 |
| 109 | Width; tip of stress bridge (for non-welded link only) | 0.65 | A8415859 |
| 110 | Guide tabs offset height; inner surface max., and outer surface min. (w/steel plug in link before retention test). | 0.65 | B7190256 |
| 111 | Width; flared end stress arm (w/ring) | 1.0 | A7187494 |
| 112 | Missing or improper detents | 1.5 | Visual |
| Minor: | | | |
| 201 | Missing or defective protective coating | 2.5 | Visual |
| 202 | Missing or improper identification | 2.5 | Visual |
| 203 | Distance between extreme inside width of body stress arms | 2.5 | B7190405 |
| 204 | Overall length; guide tab s | 2.5 | B7190249 |
| 205 | Overall length of link | 2.5 | C7190053 |
| 206 | Inside diameter of connector ring | 2.5 | A7190052 |
| 207 | Improper clearance; feeder channel guide | 2.5 | D7190256 |
| 208 | Location: front end of front guide tab (w/steel plug in link) | 2.5 | B8409284 |
| 209 | Max. overall length; longitudinal offset of guide tabs (w/steel plug in link) | 2.5 | B7190252 |
| 210 | Min. distance; front surface of ramming guide embossment to rear surface of rear guide tab longitudinal offset | 2.5 | A8415858 |
| 211 | Workmanship | 4.0 | Visual |



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4.5.2.1.2 Packaging. Unless otherwise specified in each listed defect, the packaging requirements are specified on Packaging Data Sheet P7147028.

| Categories | Characteristics | AQL |
|---|---|---|
| Major: | | |
| 101 | Improper level of packaging and packing (see procurement documents) | 1.0 |
| 102 | Incorrect or illegible marking | 1.0 |
| 103 | Inadequate cleaning and drying | 1.5 |
| 104 | Improper preservation oil application and drainage | 1.5 |
| 105 | Improper containers | 1.5 |
| 106 | Container interior contaminated with dirt, lint, or other foreign matter | 1.5 |
| 107 | Improperly oriented links in unit package | 1.5 |
| 108 | Missing volatile corrosion inhibitor treated sheets between layers of links (Levels A and B only) | 1.5 |
| 109 | Incorrect quantity of links in shipping container | 1.5 |
| 110 | Improper closure of shipping container | 1.5 |
| Minor: | | |
| 201 | Workmanship | 4.0 |

4.5.3 Testing.

4.5.3.1 Material testing. Tests required by applicable drawings and specifications shall be performed on materials used in the manufacture of links. The contractor shall furnish the Government representative certified copies of chemical analyses and reports of physical tests as may be required for each lot of each material used. The contractor shall also furnish the Government representative certification that the materials used in packaging conform to the applicable specifications.

4.5.3.2 Accelerated corrosion testing. Accelerated corrosion testing of links shall be performed in accordance with applicable drawings and specifications.

* 4.5.3.3 Hardness testing. Samples of links shall be taken from each applicable heat treat lot (see 4.5.1.1) in accordance with MIL-STD-105, using inspection level S-4 and tested for hardness requirements (see 3.2.1) using the test method specified in ASTM method E18. Acceptance shall be based on an AQL of 0.40 percent defective.

* 4.5.3.4 Decarburization testing. A sample of 5 links shall be selected at random from each heat treat lot (see 4.5.1.1) and tested for decarburization requirements (see 3.2.1) using the test method specified in 4.6.1. Failure of any link to meet the decarburization requirement shall be cause for rejection of the heat treat lot.



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* 4.5.3.5 Pull testing. Sample links shall be taken from each heat treat lot using inspection level S-3, and tested for pull requirements (see 3.2.2) using the test method specified in 4.6.2. Acceptance shall be based on an AQL of 0.40 percent defective.

* 4.5.3.5.1 Pull testing (alternative). When the alternative method of construction is used (see D7548085), sample links shall be taken from each inspection lot using inspection level S-3, and tested for pull requirements (see 3.2.2) using the test method specified in 4.6.2. Acceptance shall be based on an AQL of 0.40 percent defective.

* 4.5.3.6 Twist testing. A separate sample of 800 links from each inspection lot shall be tested for twist test requirements (see 3.2.3) using the test method specified in 4.6.3. If one or more links fail to meet the requirement, it shall be cause for rejection of the lot.

* 4.5.3.7 Cartridge retention testing. A sample of 80 links from each inspection lot shall be tested for cartridge retention (see 3.2.4) using the test method specified in 4.6.4. Acceptance shall be based on an AQL of 0.40 percent defective.

* 4.5.3.8 Free hinging testing. The 800 link sample used in paragraph 4.5.3.6 shall be tested for the free hinging requirement (see 3.2.5) using the test method specified in 4.6.5. If one or more links fail to meet the requirement, it shall be cause for rejection of the lot.

* 4.5.3.9 Weld testing. Welding shall be tested using the test method and sampling procedures specified in MIL-W-45223 or MIL-W-6858 as applicable.

4.5.3.10 Determination of cleanliness testing. Links shall be tested for cleanliness using the sampling procedure and test methods specified in MIL-P-116.

4.5.4 Inspection equipment.

* 4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified, responsiblity for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment IEL7147028 and IEL7548085 and for all other inspection equipment required to perform inspection prescribed by applicable specifications shall be in accordance with MIL-I-45607.

* 4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.



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4.6 Test methods.

* 4.6.1 Decarburization test. The cross section of links (see 4.5.3.4) shall be polished, etched with a 3-5 percent Nital solution and examined under a microscope of not less than 500 power.

* 4.6.2 Pull test. Each sample link shall be tested using the inspection equipment specified on Drawing F7190358. The links shall be inserted in the fixture and subjected to a 600 lb. load.

4.6.3 Twist test. The sample links shall be tested by assembling the links with Government standard 20mm dummy cartridges into belts of 25 linked cartridges. Each belt shall be laid out horizontally to its full length with the open side of the links down, and one end of the belt shall be flipped sharply by hand 180 degrees to its other side so that the twisting action will progressively move along the belt to the free end. The test shall be repeated with the belt positioned with the open side of the links up.

4.6.4 Cartridge retention test. The sample links shall be tested using the inspection equipment and method specified on Drawing F7190403. Partial extraction shall be classified as a failure (see 6.3.6).

4.6.5 Free hinging test. The sample links shall be tested by assembling the links with Government standard 20mm dummy cartridges into belts of 25 linked cartridges. Each belt shall be laid out horizontally to its full length, with the open side of the links up, and one end shall be drawn back upon the remainder of the belt until the belt is completely reversed (open side of links down). One end of the reversed belt shall then be drawn back upon the remainder of the belt until the belt is returned to its original (open side of links up) position.

4.6.6 Functioning test. A sample of 200 links taken from first article shall be tested by the testing agency for the functioning requirement (see 3.2.6) by assembling the links into belts using Government standard 20mm cartridges and firing the belts in bursts of 50 rounds each in an M39 type automatic gun.

5. PREPARATION FOR DELIVERY

* 5.1 Pilot pack. A pilot pack shall be forwarded in accordance with the contract (see 6.1). Pilot packs shall be packaged to the level of packaging specified in the contract and packed level C in accordance with the requirements of Packaging Data Sheet P7147028.



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* 5.2 Preparation for delivery shall be in accordance with Packaging Data Sheet P7147028 using the levels of preservation, packaging, and packing specified in the contract (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify.

6.1.1 Procurement requirements.

a. Title, number and date of this specification.
b. List of drawings and specifications pertinent to the link, showing applicable revision dates.
c. Requirements for submission of first article by the contractor (see 3.1).
d. Index of Inspection Equipment List pertinent to the link showing applicable revision dates.
e. Responsibilities for acquisition, maintenance and disposition of acceptance inspection equipment if other than specified (see 4.5.4.1).
f. Responsibility for furnishing dummy cartridges.
g. Requirements for submission of pilot pack (see 3.1 and 5.1).
h. Selection of applicable levels of preservation, packaging and packing (see 5.2).
i. Disposition instructions for link test samples.

* 6.1.2 Contract data requirements. The requirements for the records of examination and all tests will be listed directly on a DD Form 1423 incorporated into the contract.

* 6.1.3 Management control systems requirements. When warranted, the contract should specify the application of MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

* 6.1.4 Unless otherwise specified (see 6.1.1e), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

6.2 Normally, the lot number will consist of three parts - a manufacturer's identification symbol, interfix number, and a serial number, each separated by hypens. The identification symbol consists of letters assigned to indicate the identity of the manufacturer. The interfix number identifies the configuration with respect to specific design within a specific contract. The serial number identifies the lot according to its sequence of production under the conditions of manufacture represented by the interfix.



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6.3 Definition of terms.

6.3.1 Deformed links. Links whose shape or dimensions, or both, have been mutilated or altered by processing to the extent that the links are unsuitable for their intended use.

6.3.2 Cracks. See Figures 1, 2, and 3.

6.3.2.1 Cracks in principal areas. Any visible fracture of the link extending through the stock thickness in depth or a distance in length equal to or greater than the material stock thickness.

6.3.2.2 Cracks in nonprincipal areas. Any visible fracture of the link extending through the stock thickness in depth and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length.

6.3.3 Tears.

6.3.3.1 Tears in principal areas. Any visible fracture of the link with ragged or irregular edges extending through the stock thickness in depth or a distance in length equal to or greater than the material stock thickness.

6.3.3.2 Tears in nonprincipal areas. Any visible fracture of the link with ragged or irregular edges extending through the stock thickness in depth and equal to or greater than the stock thickness in length; or any visible fracture less than stock thickness in depth and greater than two times stock thickness in length.

* 6.3.4 Burs. Any rough edge on any part of the link, which in the opinion of the Government representative, would produce a definite removal of metal from a cartridge case, cause injury to personnel in handling, or result in binding of the link.

6.3.5 Principal areas. The body arms (except for the edges facing inward between the flared tips), stress bridge arms, and the connecting ring.

6.3.6 Partial extraction. When only one end of the mandrel or round becomes free.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only



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Custodians:
Army - WC

Preparing Activity:
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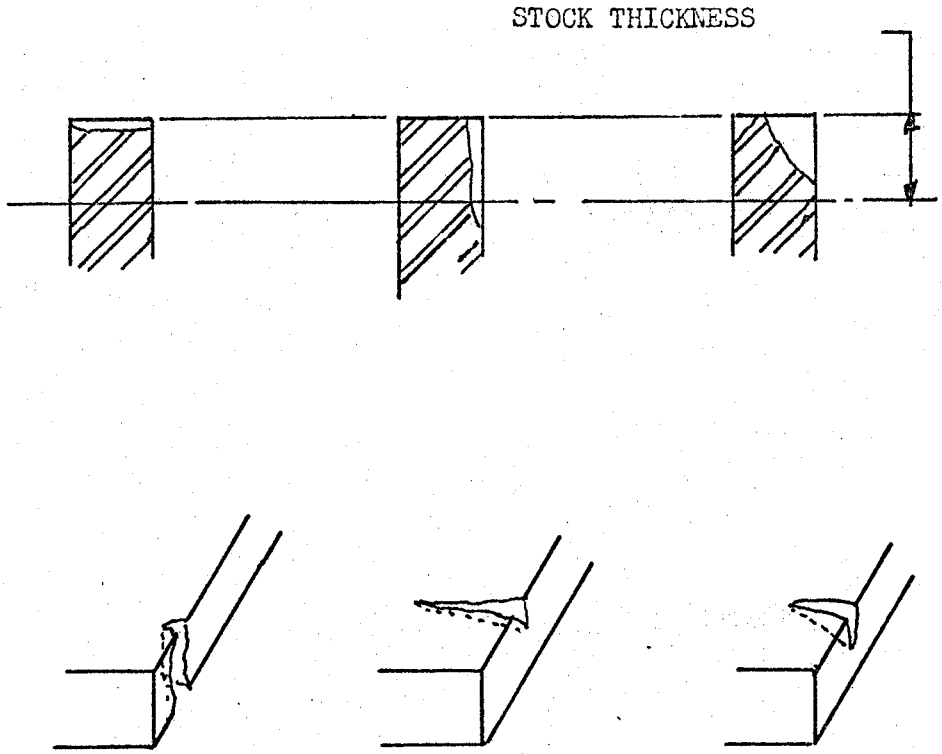
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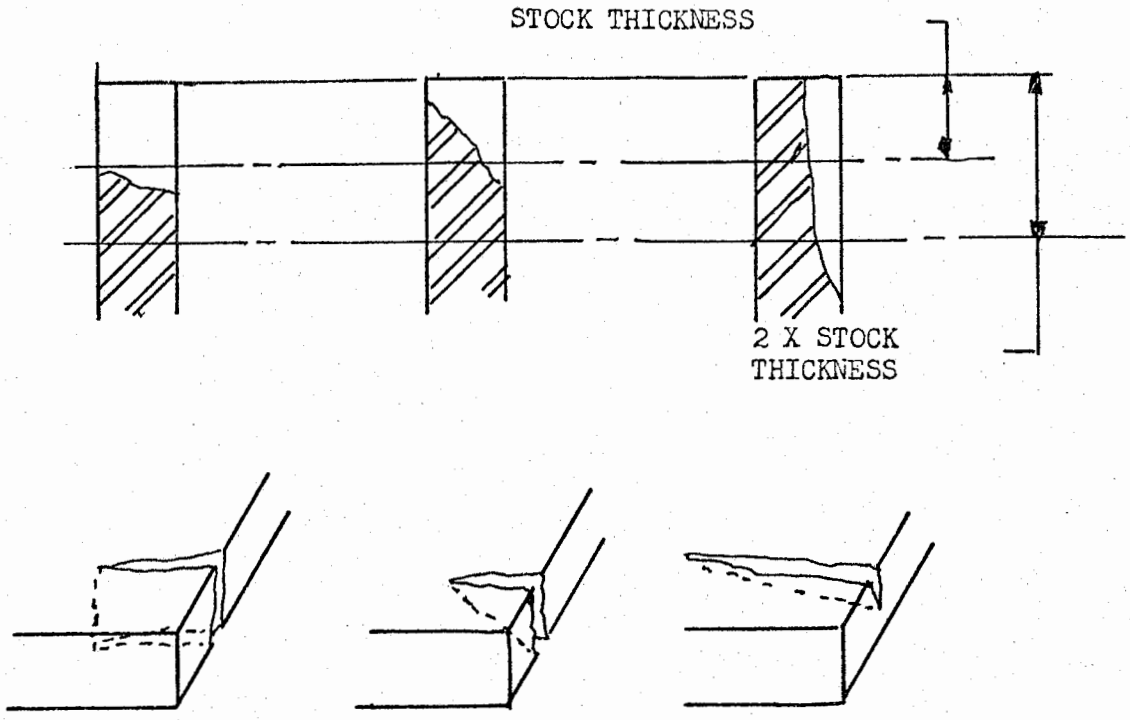



CRACKS IN PRINCIPAL AREAS

Figure 1



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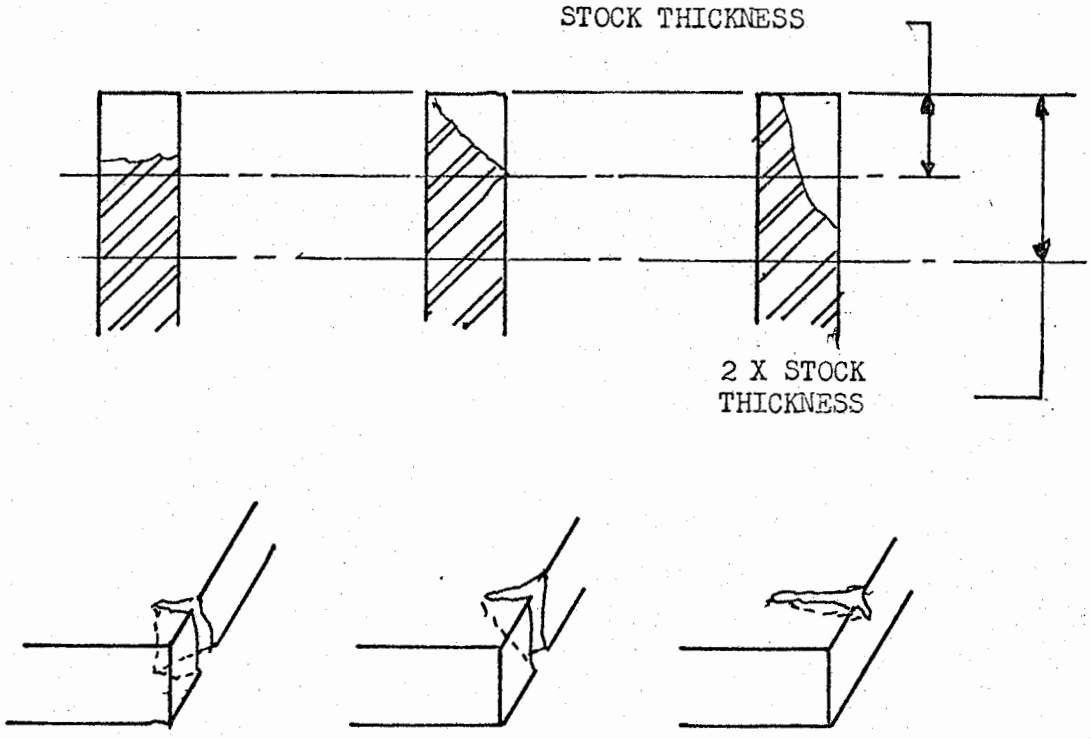

ALL NOT ACCEPTABLE

CRACKS IN NONPRINCIPAL AREAS

Figure 2



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CRACKS IN NON PRINCIPAL AREAS

Figure 3



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